

Time-dependent covariates and survival curves

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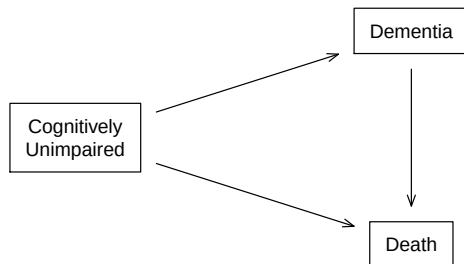
Ideas

- ▶ Time-dependent covariates are very useful
- ▶ Absolute risk and hazard ratios are complimentary
- ▶ In multistate models, both are essential
- ▶ But you can't compute $\Pr(\text{future outcome})$ with a time-dependent variable

Ideas

- ▶ Time-dependent covariates are very useful
- ▶ Absolute risk and hazard ratios are complimentary
- ▶ In multistate models, both are essential
- ▶ But you can't compute $\text{Pr}(\text{future outcome})$ with a time-dependent variable
- ▶ But I need it ...

Mayo Clinic Study of Aging



Data

- ▶ 4984 subjects, up to 16 years follow-up (median 4)
- ▶ 712 dementia events, 1852 deaths
- ▶ 51% male
- ▶ 27% APOE carrier
- ▶ initial amyloid: 24, 7, 5, 64% normal, moderate, high, NA
- ▶ initial CMC 0: 753, 1: 1005, 2: 1087, 3: 1088, 4-5: 897 6-7: 114

Amyloid by APOE and sex

M+ high

M- high

M+ moderate

M- moderate

M+ normal

M- normal

F+ high

F- high

F+ moderate

F- moderate

F+ normal

F- normal

4y greater education

2 add'l CMC

Without dementia
to dementia

Amyloid

High

Moderate

Normal

Sex

Male

Female

APOE e4 genotype

Carrier

Non-carrier

4y greater education

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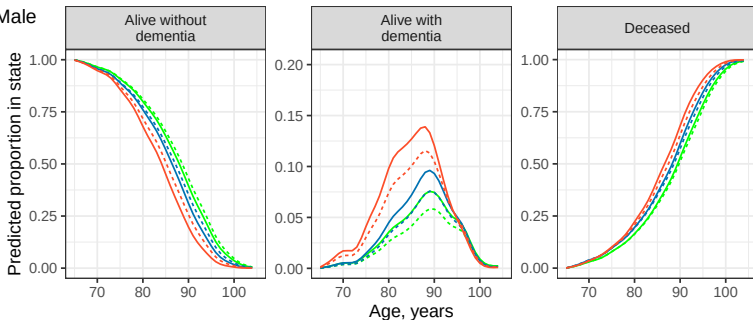
2 add'l CMC

Dementia
to death

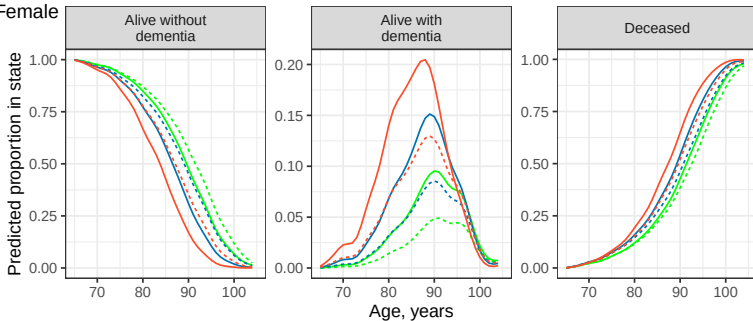
Hazard ratio (95% confidence interval)

Amyloid — normal — moderate — high APOE — carrier non-carrier

A. Male



B. Female



The problem with “baseline” CMC

- ▶ Mean fu is 8 years
- ▶ Create the curve for a 65 year old, low baseline CMC
 - ▶ by age 73 all new subjects
 - ▶ by age 81 again new
 - ▶ real CMC drifts up, current cohort CMC does not
- ▶ The usual KM will be biased, too few events

Expected Survival (survexp)

- ▶ Cohort of 85 subjects, what is their expected 10 year survival?
- ▶ Sex, age in days, enrollment date, $p_i(0) = 1$
 1. look up 85 daily hazards in US rate tables
 2. $h(t) = \sum_{i=1}^{85} p_i(t)h_i(t) / \sum p_i(t)$
 3. $p_i(t) = p_i(t) \exp(-h_i(t))$
 4. move ahead one day
- ▶ $S = e^{h_1} e^{h_2} \dots$

MCSA

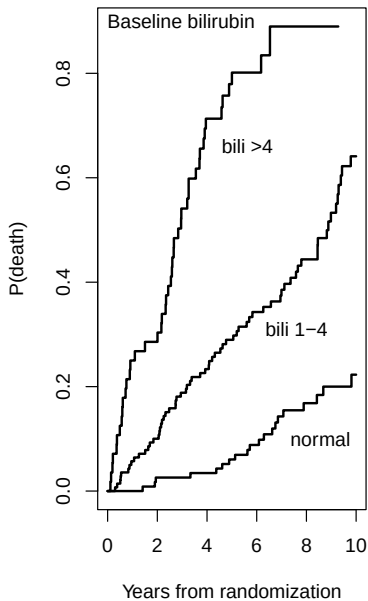
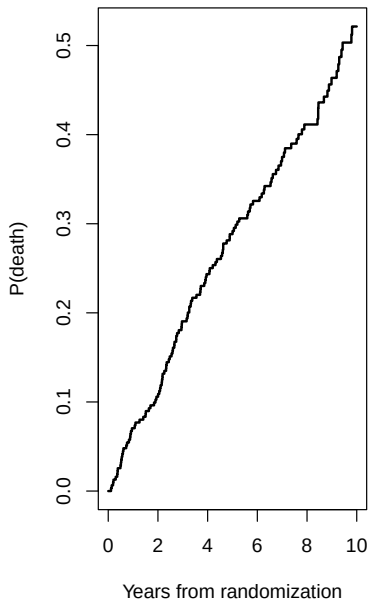
- ▶ Use modified MSH to create a 3 column rate table by age, sex, APOE, amyloid, CMC (ND:dem, ND:death, dem:death)
- ▶ AJ estimates of future CMC/death, given start at 0, start at 1, ...
- ▶ AJ estimates of future amyloid/death, given start low, start mod, start high
- ▶ starting point π_i of state * CMC * amyloid
 1. take weighted average across p_i of the hazard, for ND:dem, ND:death, dem:death
 2. plug these into an $H(t)$ matrix
 3. Update the π vectors for each subject, move ahead 1 year
- ▶ $p(t) = e^{H_1} e^{H_2} \dots$
- ▶ “distribution of recruited individuals”

Primary Biliary Cirrhosis

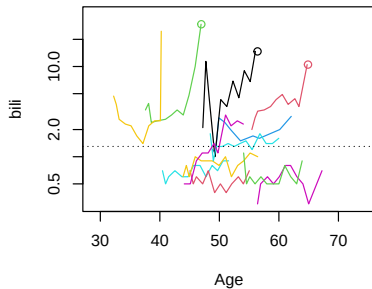
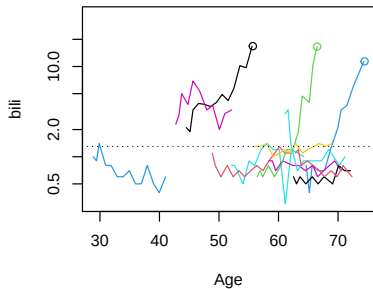
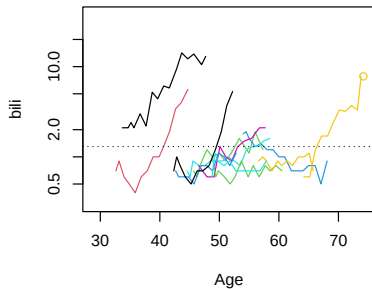
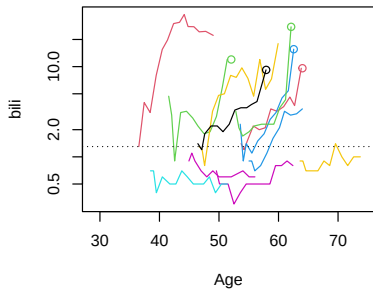
- ▶ Two clinical trials of D-penicillamine for treatment of PBC
- ▶ No treatment effect
- ▶ Data merged, and used as a model for natural history of PBC (n=418)
- ▶ Covariates of bilirubin (10.4), age (5.1), albumin (3.8), edema (3.3), prothrombin time (3.1)
- ▶ Concordance = .835
- ▶ pbc and pbcseq data sets

Model

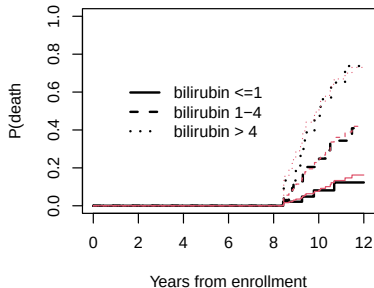
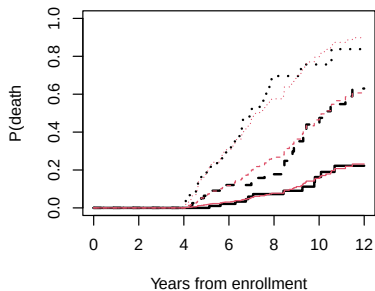
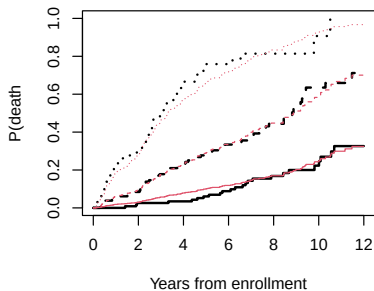
- ▶ Use only bilirubin and age, categorical bilirubin
- ▶ Use pbcseq subset ($n=312$)
- ▶ Planned visits at 6m, 1yr, yearly thereafter
- ▶ Time-dependent bilirubin



	Hazard Ratio			
	Age10	bili 1–4	bili > 4	C
Time-fixed	1.5	3.5	14.0	0.79
Time-dependent	1.6	2.9	27.1	0.86



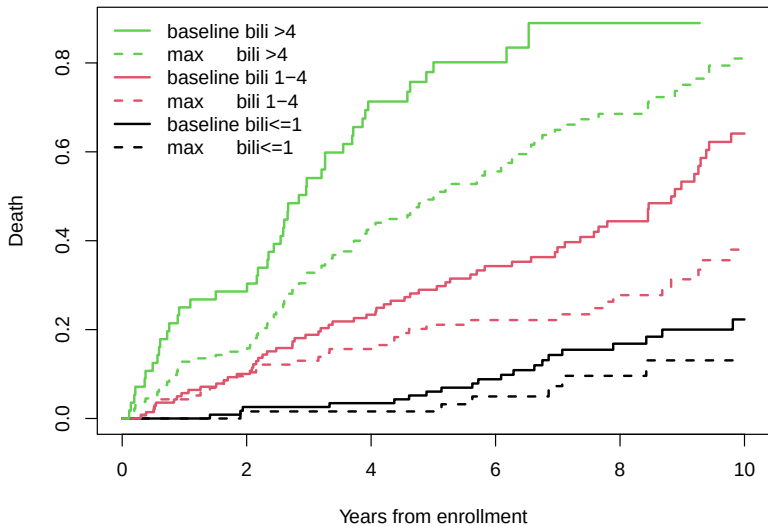
Conservative approach



- ▶ No explicit use of time-dependent data
- ▶ Curves are correct, but meh
- ▶ Aside: these are marginal over age. This matters.

Worst approach

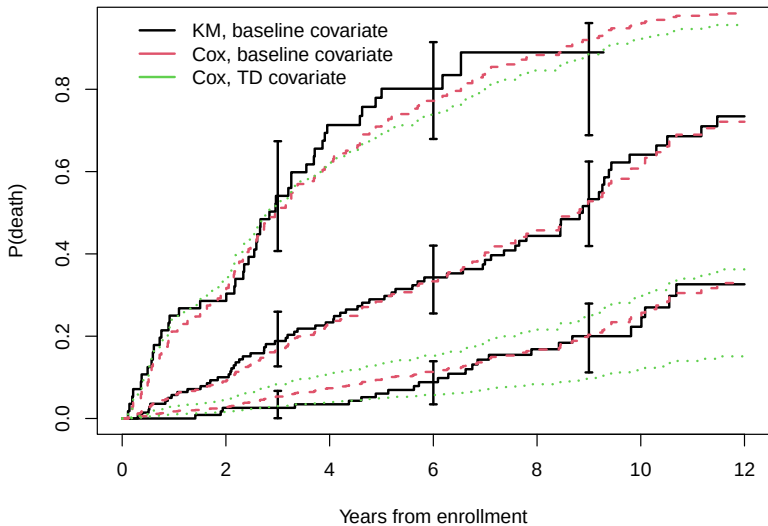
- ▶ Categorize each subject by their max bilirubin achieved
- ▶ A variant of immortal time bias
 - ▶ Covariate that depends on the future
 - ▶ Endpoint that depends on the future
 - ▶ Selection that depends on the future



Useless: static referrent

- ▶ Fit the time-dependent Cox model
- ▶ Predict the survival for a fixed covariate set
- ▶ Example in Therneau and Grambsch

```
fit <- coxph(Surv(year1, year2, death) ~ age10 + bili3,  
            pdata, id=id)  
dummy1 <- data.frame(age10= mean(pdat0$age10),  
                    bili3= "1-4")  
surv1 <- survfit(fit, dummy1)  
  
dummy2 <- data.frame(age10= pdat0$age10,  
                    bili3="1-4")  
surv2 <- survfit(dummy2)
```



What went wrong?

- ▶ Baseline covariates
 - ▶ Cox model based on baseline bilirubin
 - ▶ Prediction for someone with a specified baseline bilirubin
- ▶ TD covariates
 - ▶ Cox model based in TD bilirubin
 - ▶ Middle curve is the prediction for a cohort of subjects who start with bilirubin 1-4, and then *never change*
- ▶ Baseline bili: 116 normal, 140 1-4, 56 > 4
- ▶ Almost all of the 140 progress

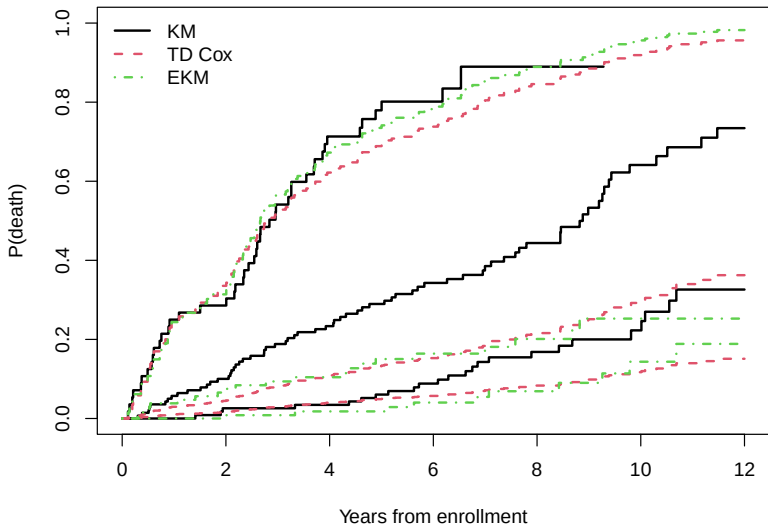
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 - ▶ Middle curve is the prediction for a cohort of subjects who start with bilirubin 1-4, and then *never change*
- ▶ Baseline bili: 116 normal, 140 1-4, 56 > 4
- ▶ Almost all of the 140 progress
- ▶ Correct estimate for a cohort which doesn't exist

Useless: Extended Kaplan-Meier

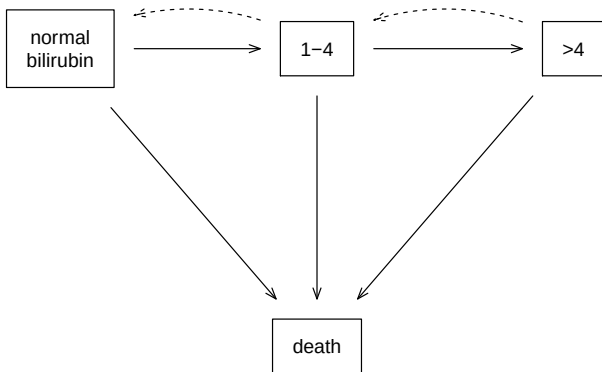
- ▶ Snappin et al, American Statistician, 2005
- ▶ Use simple survfit on the time-dependent data

```
snappin <- survfit(Surv(year1, year2, death) ~ bili3,  
                   data= pdata)
```

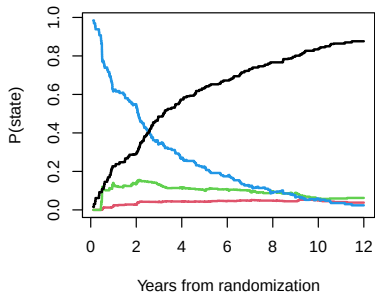
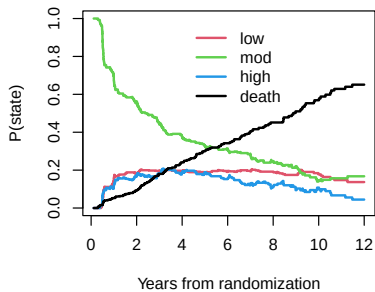
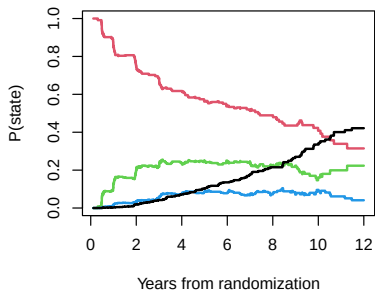
Big picture

- ▶ To predict future survival with a TD covariate one needs to specify a *covariate path*
- ▶ Possible in some cases
- ▶ Allowed by the software
- ▶ Alternative: use a multistate model

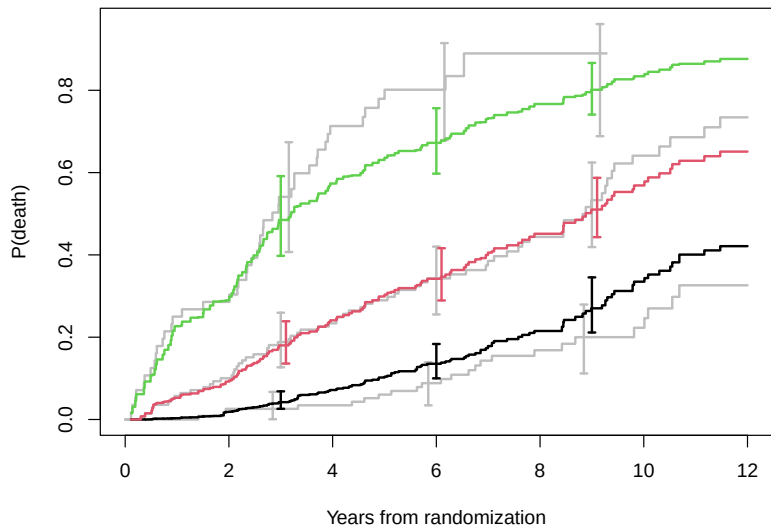


Multistate curves

- ▶ Ordinary Cox model: specify the covariates
- ▶ AJ: specify starting time and starting state
- ▶ MSH: specify covariates, starting time, starting state



```
aj3  <- survfit(Surv(year1, year2, bstate) ~ 1, pdata,  
                istate=bili3, id=id)  
aj3a <- survfit(Surv(year1, year2, bstate) ~ 1, pdata,  
                istate=bili3, p0=c(1,0,0,0), id=id)  
aj3b <- survfit(Surv(year1, year2, bstate) ~ 1, pdata,  
                istate=bili3, p0=c(0,1,0,0), id=id)  
aj3c <- survfit(Surv(year1, year2, bstate) ~ 1, pdata,  
                istate=bili3, p0=c(0,0,1,0), id=id)
```



Math

The death curve for group 1 = “normal” will have an increment at time t of

- ▶ KM: $KM(t^-) * P(\text{death at } t | \text{started in 1})$
- ▶ AJ : $\sum_{j=1}^3 P(\text{currently in } j | \text{started in 1}) P(\text{death at } t | \text{currently in } j)$
- ▶ The AJ uses all the data for all the curves.
 - ▶ smaller variance
 - ▶ Markov assumption

Hazard models

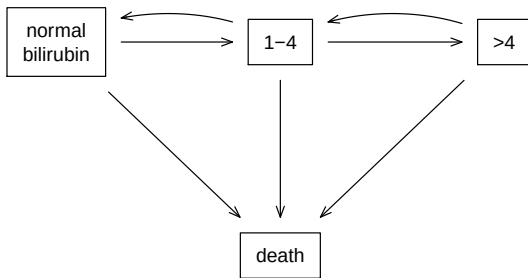
$$\lambda_{ijk}(t) = \lambda_{jk}(t) \exp(X_i \beta)$$

$$\lambda_{j4} = \lambda_d \exp(\gamma_j)$$

- ▶ A covariate now belongs to the state rather than to the subject.
- ▶ Subtle, but with software implications
- ▶ Do we want covariates for the within bilirubin transitions?

Hazard models

```
mfit0 <- coxph(Surv(year1, year2, death) ~ age10 + bili3,  
              data= pdata, ties='breslow')  
mfit1 <- coxph(list(Surv(year1, year2, bstate) ~ 1,  
                  0:4 ~ age10 / common + shared),  
              data= pdata, istate=bili3, id=id)  
  
coef(mfit0)  
      age10  bili31-4  bili3>4  
0.4780539 1.0517207 3.2988262  
coef(mfit1)  
      age10 ph(2:4/1:4) ph(3:4/1:4)  
0.4780539  1.0517207   3.2988262
```



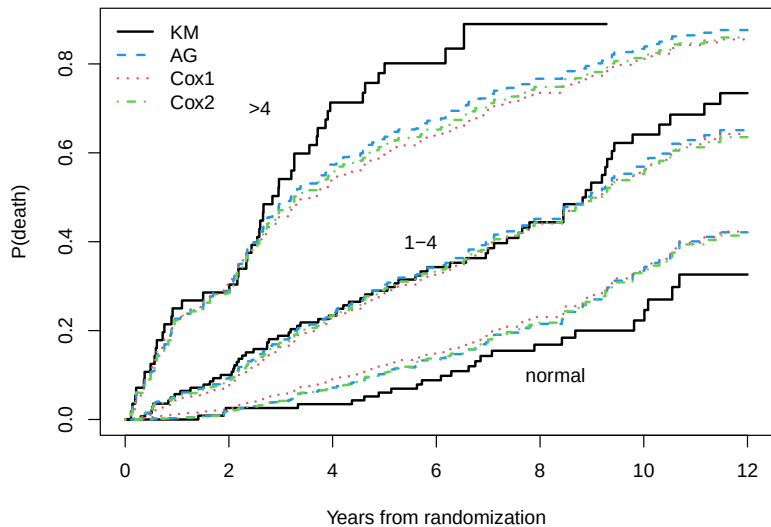
$$\lambda_{14} = \lambda_d \exp(\beta age) \quad (1)$$

$$\lambda_{24} = [\lambda_d \exp(\gamma_{12})] \exp(\beta age) \quad (2)$$

$$\lambda_{34} = [\lambda_d \exp(\gamma_{13})] \exp(\beta age) \quad (3)$$

$$(4)$$

- ▶ Same likelihood as the simple TD model, wrt death
- ▶ But now we can estimate curves properly
- ▶ γ coefficients belong to the hazard/state pair and not the subject



Open questions

- ▶ Multistate AJ: honest curves, but no $\hat{\beta}$ for bilirubin
 - ▶ How critical is the Markov assumption?
 - ▶ Is the reduction in variance bankable?
 - ▶ How many sub-states?
 - ▶ Multiple variables?
- ▶ Multistate hazard model: correct curves AND coefficients
 - ▶ How many sub-states?
 - ▶ Multiple variables (additivity)
 - ▶ Constraints
 - ▶ Code
 - ▶ Variance (IJ)